Kaustav Mitra

Curriculum Vitae *[abridged version]*

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Education

- **Ph.D.**, *Yale University*, thesis advisor: Frank C. van den Bosch — 2025
- **M.Sc. & M.Phil., Astrophysics**, *Yale University*, Ph.D. coursework: [link] — 2021
- **B.Sc. & M.Sc., Physics**, *Presidency University (Kolkata, India)*, coursework: [link] — 2018

Selected Talks ($\diamond\diamond \implies$ *invited talk*)

- Variant of: **"Differentiable forward-model of galaxy and halo assembly meets modern survey data"**
  - ◇◇ MockNYC Conference, Flatiron Institute / Simons Foundation, New York (USA). — Jan 2026
  - – Early Career Scientist Symposium, High Energy Phys., Argonne National Lab (USA). — Jan 2026
- Variant of: **"Dynamical modelling of satellite galaxies to infer galaxy-halo connection and cosmology"**
  - ◇◇ Astronomy Seminar, Department of Physics & Astronomy, Clemson University, (USA). — Feb 2026
  - – AAS Meeting (247th), Phoenix, AZ (USA). — Jan 2026
  - ◇◇ Galaxy Formation Talk, Academia Sinica (ASIAA), Taipei City (Taiwan). — Feb 2025
  - ◇◇ KIAA-PKU Seminar, Kavli Institute for Astronomy and Astrophysics, Beijing (China). — Dec 2024
  - – Santa Cruz Galaxy Workshop, University of California, Santa Cruz (USA). — Aug 2024
  - – N3AS Summer School in Multi-Messenger Astrophysics, UCSC, California, (USA). — Aug 2022
  - – 53^{rd} annual meeting, Division on Dynamical Astronomy (DDA), CCA, NY (USA). — April 2022
- Variant of: **"Disk-Jet connection in blazars: modelling radiative transfer in multi-zone turbulent jets"**
  - – Advances in Astroparticle Physics & Cosmology, SINP, WB (India). — Jan 2020
  - – The 29^{th} New England Regional Quasar/AGN Meeting, MIT Kavli Institute, MA (USA). — May 2019
- Variant of: **"Halo Occupation of Quasars: AGN Unification From a Cosmological Perspective"**
  - – APS April meeting (Distinguished Student Fellowship), Columbus, OH (USA). — April 2018
  - – 36^{th} meeting, Astronomical Society of India, Hyderabad (India). — Feb 2018

Teaching Experience

At Yale University *(see the long-version CV for past teaching experiences)*

- **Teaching Assistantship in undergraduate courses**
  - Galaxies and the Universe (ASTR 120) *main instructor:* Robert Zinn
  - Introduction to Observational Astronomy (ASTR 155) *main instructor:* Michael Faison
  - Planets and Stars (ASTR 110) *main instructor:* Robert Zinn
  - Galaxies and the Universe (ASTR 120) *main instructor:* Jeffrey Kenney
  - Origins and the Search for Life in the Universe (ASTR 130) *main instructor:* Debra Fischer

Scholarships and Awards

1. 3 consecutive times awardee of Yale Dean's Colloquium fund, as the organiser of Yale cosmology seminar series. — 2022, 2023, 2024
2. 2^{nd} prize winner of Yale 3-minute thesis (3MT) competition in the physical sciences division — 2023
3. American Physical Society Distinguished Student Fellowship — 2018

4. S. N. Bhatt Memorial Excellence Fellowship, for undergraduate summer research, at the International Centre for Theoretical Sciences (ICTS-TIFR), India. — June - Aug, 2017
5. Kishore Vaigyanik Protsahan Yojana (KVPY) scholarship, Department of Science and Technology, Government of India. — 2013 - 2018
6. DST-INSPIRE Fellowship, Department of Science and Technology, Government of India. — 2013
7. Jagadis Bose National Science Talent Search (JBNSTS) scholarship. — 2013

Outreach

Public Talks

- Science In The News (Talks in public libraries, partnering with lifelong-learners' programs) — 2024
  "**Ray tracing from computers to the cosmos**" (jointly with 2 computer science PhD students)
  - Essex library, 33 West Avenue, Essex, CT 06426 — May 11th
  - Mark Twain library, 439 Redding Road, Redding, CT 06896 — May 21st
  - Norwich public library, 368 Main St. Norwich, VT 05055 — May 30th
  - Schiller Shoreline Institute for Lifelong Learning, Nathanael B. Greene Community Center 32 Church St, Guilford, CT 06437 — May 28th
- **"Using high-school physics to unlock the strangest mysteries of the universe"** (some variation of this)
  - 2 pedagogical classes for high school students from Connecticut (Yale Splash initiative). — Nov 2023
  - 1 hour lecture, addressing high school students of St. Timothy's School, Maryland. — March 2021
  - Yale Pathways to Science for high school students from Connecticut. (20 minute talk) — June 2020
  - Open Labs Exploring Science week for middle and high school students of greater New Haven area. — Feb 2020
- "What are stars made of?", Open Labs Science Cafe talk for middle and high school students. — Oct 2020
- "How to weigh galaxies and why do we need to?", Astronomy on Tap, New Haven, CT (USA). — Feb 2020

Leitner Family Observatory and Planetarium

- I have regularly presented planetarium shows at Yale's Leitner Family Observatory and Planetarium, and have led numerous stargazing sessions there during the public-nights, every Tuesdays. — 2019, 2020

Professional Commitments and Leadership Roles

- Judge: Chambliss Astronomy Achievement Student Awards, American Astronomical Society. — 2026
- Mentor on Project SHORT: a student-led non-profit organisation guiding students from underprivileged backgrounds, mostly from developing countries, on graduate school applications. — 2022 - Now
- I started the weekly Yale Cosmology Seminars and I was the main organizer for 2 years. — 2022 - 2024
- PhD student interviewer in graduate student recruitment. — 2022 - 2024
- Content Editor, Yale Department of Astronomy Newsletter. — 2023
- Co-organizer of weekly Galaxy Lunch seminars, Yale Department of Astronomy. — 2022 - 2023
- Graduate student representative in the prospective student webinar and information session. — Nov 2022
- Served in the student panel for interviewing and evaluating shortlisted applicants in the departmental faculty hiring process. — 2022
- Two times "Data tsar" at CHIME telescope (one responsible for monitoring the smooth running of the telescope) — 2019 - 2020
- Student member of the American Physical Society and member of American Physical Society (APS) Forums On Graduate Student Affairs (FGSA) and Outreach and Engaging the Public (FOEP). — 2017 - 2019
- Science communication and public outreach volunteer American Physical Society (APS) April meeting, Columbus, Ohio, USA. — April 2018
- Member of scientific organising committee and session chair of astrophysics session Undergraduate physics symposium, Presidency University. — April 2017
- Student Coordinator, 'Advanced School on Gravitational Waves', Presidency University — December 2016

List of Publications

Refereed Papers

1. **Kaustav Mitra**, Frank C. van den Bosch, Josephine Baggen, Johannes U. Lange, (in rev.) "BASILISK IV. No S_8 Tension with Satellite Kinematics". OJAp [in review]
2. **Kaustav Mitra**, Frank C. van den Bosch, (in rev.) "BASILISK III. Stress-testing the Conditional Luminosity Function model". OJAp [in review]
3. Josephine Baggen, Frank C. van den Bosch, **Kaustav Mitra**, (in rev.) "The Impact of Baryonic Effects on the Dynamical Masses Inferred Using Satellite Kinematics". OJAp [in review]
4. J. Sebastian Monzon, Frank C. van den Bosch, **Kaustav Mitra** (2024). "Constraining the low mass end of the stellar halo mass relation with surveys of satellite galaxies". ApJ, Volume 976, Issue 2, id.197, 23 pp.
5. **Kaustav Mitra**, Frank C. van den Bosch, Johannes U. Lange, (2024) "BASILISK II. Improved constraints on the galaxy-halo connection from satellite kinematics in SDSS". MNRAS, 533(3), 3647-3675
6. Aritra Ghosh, et al., *including* **Kaustav Mitra** (2024). "Denser environments cultivate larger galaxies: a comprehensive study beyond the Local Universe with 3 million Hyper Suprime-Cam galaxies" 971(2), id.142, 23 pp.
7. Pieter van Dokkum, et al., *including* **Kaustav Mitra** (2023). "A Candidate Runaway Supermassive Black Hole Identified by Shocks and Star Formation in its Wake". ApJL, 946(2), 14 pp.
8. Saugata Barat, Ritaban Chatterjee, **Kaustav Mitra** (2022). "Locating the GeV emission region in the jets of blazars from months time-scale multiwavelength outbursts". MNRAS, 515(2), 1655-1662.
9. Aritra Kundu, Ritaban Chatterjee, **Kaustav Mitra**, Sripan Mondal (2022). "RMS-flux relation and disc-jet connection in blazars in the context of the internal shocks model". MNRAS, 510(3), 3688-3700.
10. Anwesh Majumder, **Kaustav Mitra**, Ritaban Chatterjee, C. Megan Urry, Charles Bailyn, Prantik Nandi (2019). "Physical inference from the γ-ray, X-ray, and optical time variability of a large sample of Fermi blazars". MNRAS, 490(1), 124-134.
11. Sagnick Mukherjee, **Kaustav Mitra**, Ritaban Chatterjee (2019). "The accretion disc-jet connection in blazars". MNRAS, 486(2), 1672-1680.
12. **Kaustav Mitra**, Suchetana Chatterjee, Michael A. DiPompeo, Adam D. Myers, Zheng Zheng (2018). "The Halo Occupation Distribution of obscured quasars: revisiting the unification model". MNRAS, 477(1), 45-55.

Conference Proceedings

1. **Kaustav Mitra**, Frank C. van den Bosch, Josephine Baggen, Johannes U. Lange (2025). "Novel probe of S8 tension: insensitive to halo assembly bias and incorporating baryonic effects". 245th Meeting of the American Astronomical Society, id. 250.25. Bulletin of the American Astronomical Society, Vol. 57, No. 2.
2. **Kaustav Mitra**, Frank C. van den Bosch (2022). "Dynamical modelling of satellite galaxies to infer galaxy-halo connection and cosmology". AAS Division on Dynamical Astronomy meeting #53, id. 405.04. Bulletin of the American Astronomical Society, Vol. 54, No. 4.
3. Adam D. Myers, Michael A. DiPompeo, **Kaustav Mitra**, Ryan C. Hickox, Suchetana Chatterjee, Kelly Whalen (2018). "Characterizing the evolution of WISE-selected obscured and unobscured quasars using HOD models". American Astronomical Society, AAS Meeting #232, id. 322.01.
4. **Kaustav Mitra**, Suchetana Chatterjee, Michael A. DiPompeo, Adam D. Myers, Zheng Zheng (2018). "Halo Occupation of Quasars: AGN Unification From a Cosmological Perspective". American Physical Society (APS) April Meeting 2018, abstract id.S16.001